

CSE 400: Fundamentals of Probability in Computing

Lecture 18: Marginal PMF, Correlation, Covariance, Joint Gaussian RVs

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1 Marginal PMF

1.1 Definition

The marginal PMF of a random variable is obtained by summing the joint PMF over all possible values of the other variable.

$$P_X(x) = \sum_y P_{X,Y}(x, y), \quad P_Y(y) = \sum_x P_{X,Y}(x, y)$$

1.2 From Joint PMF Table

To compute marginal PMF:

- Sum across rows to get $P_X(x)$
- Sum across columns to get $P_Y(y)$

1.3 Key Idea

Marginalization removes dependence on the other variable.

2 Correlation

2.1 Definition

Correlation is measured using the product moment:

$$R_{XY} = \mathbb{E}[XY]$$

For continuous variables:

$$\mathbb{E}[XY] = \int \int xy f_{X,Y}(x, y) dx dy$$

2.2 Interpretation

- Measures average product of X and Y
- Depends on scale of variables
- Computed about origin (not mean)

2.3 Limitation

- Mixes mean, spread, and dependence
- Not a clean measure of dependence

Special Case:

$$R_{XY} = 0 \Rightarrow \text{Orthogonal about origin}$$

3 Covariance

3.1 Definition

Covariance measures dependence after removing means:

$$\text{Cov}(X, Y) = \mathbb{E}[(X - \mu_X)(Y - \mu_Y)]$$

3.2 Equivalent Form

$$\text{Cov}(X, Y) = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y]$$

3.3 Interpretation

- Positive $\Rightarrow$ variables move together
- Negative $\Rightarrow$ variables move oppositely
- Zero $\Rightarrow$ no linear relationship (uncorrelated)

3.4 Properties

1. **Symmetry:**

$$\text{Cov}(X, Y) = \text{Cov}(Y, X)$$

2. **Variance as Special Case:**

$$\text{Cov}(X, X) = \text{Var}(X)$$

3. **Linearity:**

$$\text{Cov}(aX + b, Y) = a \cdot \text{Cov}(X, Y)$$

4. **Independence:**

$$X \perp Y \Rightarrow \text{Cov}(X, Y) = 0$$

(Converse is not always true)

4 Pearson Correlation Coefficient

4.1 Definition

$$\rho_{XY} = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y}$$

4.2 Properties

- Unit-free
- Range: $-1 \leq \rho \leq 1$
- Measures strength of linear relationship

5 Joint Gaussian Random Variables

5.1 Definition

A pair (X, Y) is jointly Gaussian if:

- Joint distribution is bivariate normal
- Marginals are Gaussian

5.2 PDF Form

$$f_{X,Y}(x, y) = \frac{1}{2\pi\sigma_X\sigma_Y\sqrt{1-\rho^2}} \exp\left(-\frac{1}{2(1-\rho^2)} \left[\frac{(x-\mu_X)^2}{\sigma_X^2} + \frac{(y-\mu_Y)^2}{\sigma_Y^2} - \frac{2\rho(x-\mu_X)(y-\mu_Y)}{\sigma_X\sigma_Y} \right]\right)$$

5.3 Interpretation

- μ_X, μ_Y control center
- σ_X, σ_Y control spread
- ρ controls dependence

5.4 Applications

- Signal processing
- Communication systems
- Modeling noisy measurements

6 Examples (Conceptual)

6.1 Covariance Computation

- Compute $\mathbb{E}[X], \mathbb{E}[Y], \mathbb{E}[XY]$
- Use:

$$\text{Cov}(X, Y) = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y]$$

6.2 Joint PDF Example

- Integrate over support
- Find expectations using double integrals

7 Exam Summary (Important)

- **Marginal PMF:** Sum out the other variable
- **Correlation:** $E[XY]$ (raw, not centered)
- **Covariance:** Measures linear dependence
- **Key Formula:**

$$\text{Cov}(X, Y) = E[XY] - E[X]E[Y]$$

- **Independence $\Rightarrow$ Zero Covariance**

- **Pearson Coefficient:**

$$\rho = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y}$$

- **Joint Gaussian:**

- Fully defined by mean, variance, correlation
- Most important continuous model